

OUSSAMA ALOUCHE

Full-Stack Engineer & Frontend Architect

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WORK EXPERIENCE

Design Systems & Frontend Experience Lead

BuildwellAI

Sept 2025 – Present

Remote (London, UK)

- Owned the complete UI/UX function as the sole designer and frontend engineer, establishing the brand identity, design system, and visual consistency across BuildwellEYE, BuildwellTHREAD, BuildwellCHAT, and BuildWell Inspect.
- Standardized UI components, colors, typography, and design patterns in Figma, ensuring a consistent experience across all products and platforms.
- Architected and built an AI-powered news platform from concept to production, integrating automated article generation, structured formatting, and dynamic image sourcing.
- Engineered the BuildWell Inspect property app with bug documentation, photo capture, structured report generation, and iterative UX improvements through research and prototyping.

AI Solutions & Fullstack Engineer

TECHNIQ8 (formerly AZ HUB — company rebrand)

May 2024 – Sept 2025

Rabat, Morocco

- Led frontend architecture and design systems across multiple client products, building scalable component libraries and delivering end-to-end Figma systems (colour tokens, typography, documentation).
- Architected backend systems with FastAPI and Flask, including a real-time speech-to-speech AI conversion pipeline and NLP-based content classification features.
- Delivered two full-stack platforms end-to-end: FOSTP (multi-step forms, real-time data sync) and RMA-connect, an IoT-based EV charging station management system.

Software Engineering Intern

ARK-X Talent Factory

Jan 2024 – Jun 2024

Rabat, Morocco

- Completed an intensive MERN stack programme; contributed to 3 cross-functional team projects using Agile methodologies and built reusable component libraries with REST API integrations.

PROJECTS

BuildwellAI Competency Ledger — Master's Thesis Project

Software Engineer & System Architect

Mar 2026 – Jul 2026

Remote / Kénitra, Morocco

- Designed and implemented a web-based platform that digitises the full lifecycle of construction competency verification to comply with the Building Safety Act 2022.
- Engineered a three-tier web application using a Next.js client, a TypeScript/Express API, and a PostgreSQL database.
- Architected the system with role-based access control, JSON Web Token sessions, federated sign-in, and transactional email capabilities.
- Built end-to-end workflows allowing operatives to claim a competency against a recognised scheme standard, submit supporting evidence or sit a knowledge assessment, and receive independent verification.
- Developed a live compliance matrix feature that allows project managers to assemble site teams and instantly compare the competencies a project requires against those the crew holds.
- Established a continuous-delivery pipeline, deploying the Next.js frontend on Vercel's content-delivery network and the stateless Express backend in a Docker container behind an Nginx reverse proxy on a Linux virtual private server.
- Integrated a tamper-evident audit log using database-level hash chaining to ensure the integrity of verification decisions.
- **Tech:** Next.js, TypeScript, Express.js, PostgreSQL, Docker, Nginx, JWT, OAuth, Vercel, Linux VPS.

Confidential Frontend Design System — NDA Freelance Project

Freelance Frontend Engineer (NDA)

Feb 2026 – Apr 2026

Remote (Polish Client)

- Engineered a comprehensive, scalable frontend design system for a confidential Polish technology company under a strict Non-Disclosure Agreement.
- Developed highly reusable component libraries, standardizing typography, color tokens, and interactive UI elements across the client's product suite.
- Ensured cross-platform visual consistency and accessibility compliance, accelerating the client's frontend development lifecycle and reducing design-to-code handoff time.
- Professional reference available from the project stakeholder upon request.

TRCKS — Cross-Sport Performance Analytics Platform

Lead Engineer & Product Designer (Personal Project)

Feb 2026 – Present

Remote

- Independently built TRCKS end-to-end — a marketing website, a Python/FastAPI analytics backend, and a Supabase/PostgreSQL database powering real-time sports science metrics (Kalman-filtered GPS, 7-zone speed breakdown, PlayerLoad, ACWR injury-risk flags); the companion Flutter mobile app is currently in active development ahead of a public launch.
- Trained and integrated custom PyTorch models for movement classification and performance trend prediction; built *Elis*, an AI coaching assistant powered by a self-hosted, fine-tuned Qwen 2.5 3B model (GGUF-quantized via llama.cpp), with a cloud provider kept as a swappable fallback.
- Designed the complete product UX in Figma across onboarding, dashboards, team management, and AI coach flows; architected the mobile app for an offline-first experience with background sync, JWT auth, and row-level security.
- Defined a freemium business model with Pro (€6/month), Founding Member (€29 for 2 years, capped 500 users), and Club (€35/month) tiers; implemented a Jest/Pytest test suite with >80% coverage on critical analytics modules.
- **Tech:** Flutter, React, TypeScript, FastAPI, PyTorch, Supabase, PostgreSQL, Redux, Qwen 2.5, llama.cpp, Gemini, Docker, Vercel.

BuildwellEYE — AI-Powered Property Intelligence *UI/UX Engineer & Frontend Developer (BuildwellAI)*

Sept 2025 – Present
Remote

- Designed and built the full UI for a computer-vision-based property inspection tool, creating data-rich dashboards for defect detection results and report exports.
- Owned the Figma design system for the product — component library, colour tokens, and interaction patterns — ensuring pixel-perfect implementation in React/Next.js.
- **Tech:** React, Next.js, TypeScript, Figma, Tailwind CSS.

BuildwellTHREAD & BuildwellCHAT — PropTech Communication Suite *UI/UX Engineer & Frontend Developer (BuildwellAI)*

Sept 2025 – Present
Remote

- Led UI/UX for two complementary products — a threaded project communication tool and an AI-assisted chat interface for property teams — designing conversation flows, notification systems, and role-based access views.
- Built responsive, accessible frontends in React/Next.js with real-time state updates and a consistent cross-product design language.
- **Tech:** React, Next.js, TypeScript, Figma, Tailwind CSS.

BuildWell Inspect — Property Inspection App *UI/UX Engineer & Frontend Developer (BuildwellAI)*

Sept 2025 – Present
Remote

- Designed and engineered a field-facing inspection app for property surveyors, covering structured bug logging, annotated photo capture, and automated PDF report generation.
- Conducted end-user research with surveyors to validate workflows; iterated on prototypes in Figma before implementing in React with an offline-tolerant data layer.
- **Tech:** React, Next.js, TypeScript, Figma, Tailwind CSS.

Workwhile — Talent Matching Platform *Fullstack Software Engineer (Academic Project)*

Apr 2025 – Jun 2025
Rabat, Morocco

- Built a full-featured recruitment platform connecting candidates and employers, with intelligent profile matching, application tracking pipelines, and dedicated recruiter dashboards.
- Designed a microservices architecture using Spring Boot with Redis caching, MongoDB storage, and inter-service REST communication, fully containerised with Docker for consistent deployment.
- **Tech:** React, Spring Boot, MongoDB, Docker, Redis, NGROK.

ASANADA LMS Platform *Frontend Engineer (TECHNIQ8)*

Mar 2025 – Sept 2025
Rabat, Morocco

- Designed and built the complete LMS frontend from scratch — course management, multi-type quiz engine, threaded discussion forums, learner progress tracking, and real-time notifications.
- Delivered a fully responsive design system with WCAG accessibility compliance and optimised bundle size via code splitting, achieving sub-2s load times across key pages.
- **Tech:** Laravel, React, JavaScript, Redux, SQL, Figma.

FOSTP Platform & RMA-connect EV Charging App *Fullstack Engineer & UI/UX Designer (AZ HUB)*

May 2024 – Feb 2025
Rabat, Morocco

- FOSTP: designed and delivered a high-performance web platform featuring complex multi-step forms, dynamic conditional logic, and real-time data synchronisation across user sessions.
- RMA-connect: built the full frontend and backend of an IoT-based EV charging station management system, handling live device state, station availability maps, and scheduling logic.
- **Tech:** ReactJS, ExpressJS, MongoDB, Figma, Node.js.

EDUCATION

Master's in Digitalisation & Information Systems Architecture

Expected Aug 2026

ENSA Kénitra Kénitra, Morocco **Specialisation:** Systems Architecture, Digital Transformation, AI Integration, and Distributed Information Systems.

Bachelor's in Physical Sciences

Sept 2018 – Jun 2023

Faculty of Sciences, Rabat Rabat, Morocco **Specialisation:** Analytical problem-solving, quantitative reasoning, and research communication.

SKILLS

Languages & Programming: JavaScript (ES6+), TypeScript, Python, Java, PHP, Dart, HTML5, CSS3, SQL, Bash

Frontend Ecosystem: React, Next.js, Flutter, Vue.js, React Native, Tailwind CSS, Redux, Zustand, SCSS/Sass, ThreeJS, WebGL, WordPress

Backend & APIs: Node.js, Express.js, FastAPI, Flask, Spring Boot, NestJS, RESTful APIs, GraphQL, WebSockets, Microservices

System Design & Architecture: Three-Tier Architecture, Event-Driven Architecture, API Gateways, Load Balancing, Caching Strategies, Offline-First Architecture

DevOps, Cloud & Infrastructure: Docker, CI/CD Pipelines (GitHub Actions, GitLab CI), Nginx Reverse Proxy, Linux Server Administration (Debian), OVH Virtual Private Servers (VPS), Vercel Edge CDN, Git/GitHub

Databases & Storage: PostgreSQL, Supabase (Managed Auth/PostgreSQL), MongoDB, Redis, CassandraDB, Hadoop, Prisma, SQLAlchemy

Security & Identity: Role-Based Access Control (RBAC), Row-Level Security (RLS), JSON Web Tokens (JWT) & Session Rotation, Federated Sign-in (OAuth), SHA-256 Hash-Chaining & Tamper-Evident Audit Logs, TLS Encryption

AI, ML & Data Science: PyTorch, Qwen 2.5 Fine-Tuning, llama.cpp, Gemini API, NLP, Kalman-Filtered Telemetry

Testing & Quality Assurance: Jest, Pytest, React Testing Library, Flutter integration_test, Unit & Integration Testing, Test-Driven Development (TDD), Automated Quality Gates

UI/UX, Product & Process: Figma, Design Systems, Reusable Component Libraries, User Research, Rapid Prototyping, WCAG Accessibility, SaaS Monetization, Agile/Scrum Methodologies

Soft Skills & Leadership: End-to-End Product Ownership, Cross-functional Communication, Technical Documentation, Remote Collaboration, Mentoring

Languages: Arabic (Native), English (C1 – Advanced), French (B2 – Professional)